1 INTRODUCTION

Survey aims

1.1 The aims of this survey were to assess attributes of the Tweed Estuary SAC River Lamprey (*Lampetra fluviatilis*) and Sea Lamprey (*Petromyzon marinus*) spawning populations through surveys of potential spawning areas, specifically to:

a *Locate how far up the Till catchment adult river and sea lamprey are able to migrate in search of spawning habitat*

b *Record all barriers which may limit migration (giving consideration to both elevated and drought flow conditions) and also those that would be complete migration barriers in any flow*

c. *Assess the accessibility of any eel passes for use by both river and sea lamprey*

d *Provide GPS locations of suitable spawning habitat for both river and sea lamprey and its condition (to be completed during peak river lamprey spawning period)*

e *Record evidence of any spawning activity (nest counts) and sightings of spawning adult river lamprey*

f *Return during the peak sea lamprey spawning period to areas highlighted as potential sea lamprey spawning habitat and record evidence of any spawning activity (nest counts) and sightings of spawning adult sea lamprey*

The River Till

1.2 The Till is the first major tributary of the Tweed to join it in fresh water (the Whiteadder joins it in salt water in the estuary) and is very different in character from the other major tributaries (going upstream- Eden, Teviot, Leader, Gala and Ettrick) in that much of its main channel is very low gradient and has substrates of sand and flat bedrock rather than the cobbles and pebbles of the other rivers. This difference is reflected in their fisheries which are mainly for Sea-trout in the Till but for Salmon in the other tributaries – there being very little suitable spawning for salmon along the main channel of the Till, its salmonid production is concentrated in its upland burns and tributaries, which are more Sea-trout spawning habitat than Salmon. On the other tributaries, the main channels are very much salmon spawning habitat, so Salmon are produced as a higher proportion to the Sea-trout of their more upland areas.

1.3 There is therefore a very large area of sediment suitable for lamprey larvae in the Till, though as lamprey use similar substrates to salmonids for spawning, actual main channel spawning habitat is more limited in both total area and distribution. This gives the possibility that the limited number of spawning areas could be catalogued for use in future monitoring of spawning of migratory lamprey in the Till catchment

1.4 The significance of the large amount of sediment in the Till catchment for lamprey in the Tweed catchment is shown by Figure 1 which combines the results of a survey of the Tweed catchment in Scotland commissioned by SNH from the Tweed Foundation in 2004 and of the English parts of the Till catchment by English Nature in 2002. The catchment of the Till is not entirely in England, the upper half of the Bowmont / Glen, the main tributary of the Till, is in Scotland, so commissioned fish surveys of the Till (or of the Tweed) are never

for the whole. This map shows that four out of the five sites in the category of most abundance (more than 2.5 larvae per metre length of bank surveyed) were in the Till catchment.

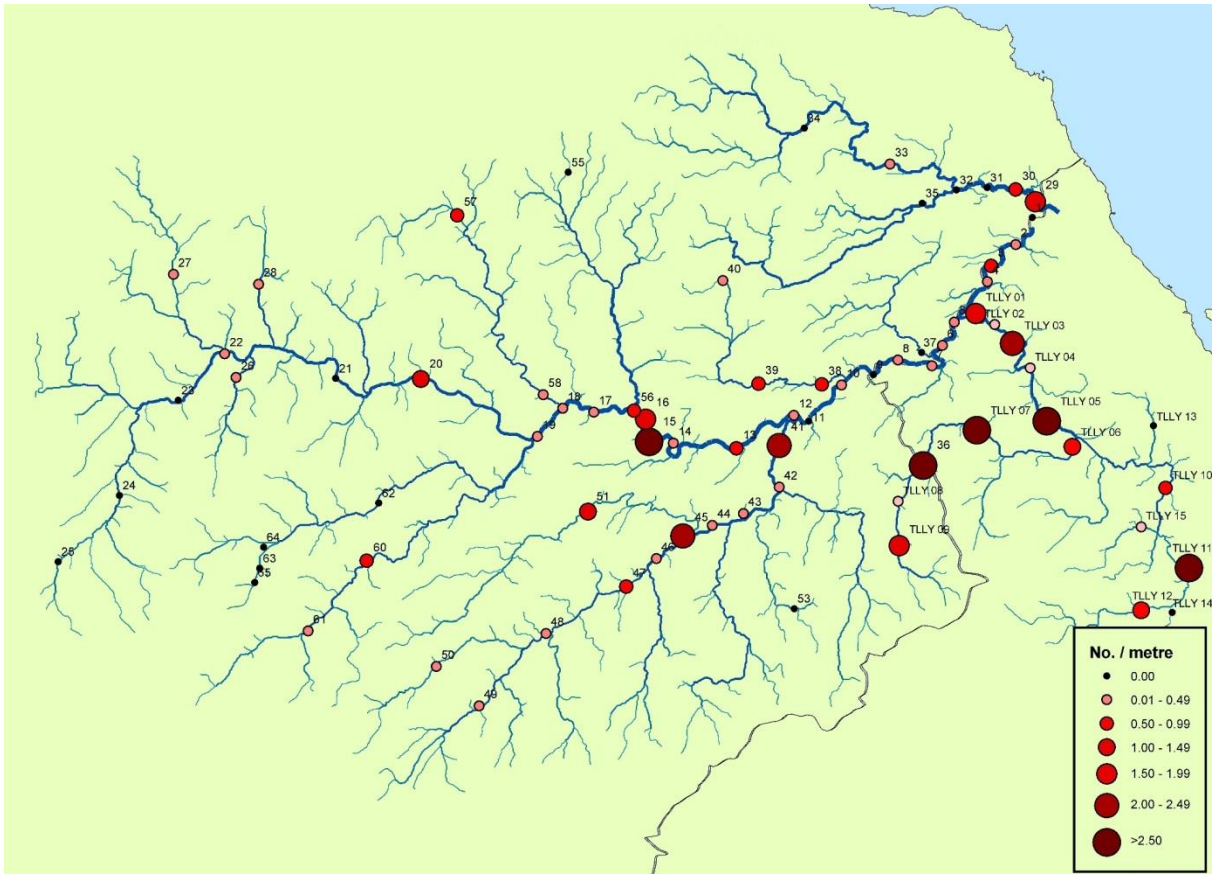


Figure 1.1 The abundance of lamprey larvae over the whole Tweed catchment. The index used is *numbers per metre length of bank* surveyed in order to combine the results of two different survey methodologies. Original data from Campbell (2002) and (2004)

1.5 A description of the Till can be found in Section 2 of the Till Restoration Strategy document on the Tweed Forum website (http://www.tweedforum.org/projects/current-projects/till_restoration_strategy).

Adult River and Sea lamprey records in and around the Till catchment

1.6 The main source of historic records for River and Sea Lamprey in the Tweed catchment is Bolam (1919). In this "*one or two*" Sea-lamprey are reported as being taken at the mouth of the Tweed every March or April (presumably in Salmon nets) and as late as the 28th of June "*at which season it is usually found higher up the river*". He also records that :

> "*Some years ago, a friend of mine who was very anxious to make a study of this Lamprey, lay in his yacht at Berwick for nearly a month without being able to secure a single specimen. This was during July, which is late for it here, but I thought we should have had no difficulty in getting one or two from some of the salmon-netting*

stations up the river. At length, an old fisherman advised my friend to go and try for them off St. Abbs Head, which he said was 'the place for them in summer'. *He went, and soon had as many as he wanted, caught on lines baited with dead fish*".

The local names recorded by Bolam were "*Lampern*" and "*Sookers*". Inquiries to local sea fishermen, made through a former salmon netsman in the Tweed estuary, Ian Bruce, by the present author did not produce any present day records of Sea Lamprey on the local coasts or in lobster pots. Interestingly, Bolam (1919) regarded the River Lamprey as being less frequently found in the Tweed than the Sea-lamprey and gives "*Barling, Cunning* and *Spanker Eel*" as being the Northumbrian names for it, though he had personally only heard the last of these in actual use.

1.7 No historic records of large and visible spawning aggregations or runs of either Sea or River Lamprey have been found for the Till so far.

1.8 The only definite recent record of adult River or Sea Lamprey in the Till catchment was made in November 2008 during a fish rescue at Kilham on the R. Glen, the main tributary of the Till. Two or three individual River Lamprey (estimated to be between 200 and 300mm) were found by R. Cussen (Natural England) and one was photographed (Plate 1.1)



Plate 1.1 River Lamprey found during a fish rescue at Kirknewton on the R. Glen on the 3rd November 2008 (Photo R Cussens, Natural England)

1.9 That both River and Sea Lamprey migrate up the main Tweed past the mouth of the Till and to a considerable distance upstream is shown by the records in Table 1.1

Table 1.1 Recent records of adult River and Sea lamprey in the Tweed catchment

SPECIES	DATE	WATERCOURSE	PLACE	NOTES
River Lamprey	1999.04.18	Leader Water	Drygrange	Smolt trap
	2000.03.29	Leader Water	Drygrange	Smolt trap. 310mm, gravid
	2000.04.06	Leader Water	Drygrange	Smolt trap. 280mm
	2004.04.29	Tweed	Coldstream	Spawning d/s of bridge
Sea Lamprey	1993 July	Jed Water	Jedburgh	Taken at the Anna Cauld
	2000.05.12	Tweed	Mertoun	Part eaten by Otter. Gravid
	2000.06.04	Ettrick Water	Selkirk	In Philiphaugh Fish Farm
	2004 end May	Tweed	Mertoun	Redds recorded
	2009.06.03	Tweed	Wark	Moribund male
	2010.06.08	Tweed	Melrose	Seen spawning

It should be noted that all these records are from upstream of at least one large instream structure with either a slap or a fish pass in it and the Ettrick and Leader records are upstream of seven (Tweed Mill, Coldstream Bridge, The Lees, Hendersyde, Kelso, Rutherford and Mertoun).

1.10 There are also records of River Lamprey in the stomachs of Goosanders (*Mergus merganser*) from the general area of the Lower Tweed (Marquiss & others, 1998), which could be associated with the Till catchment.

Table 1.2 Records of adult lamprey in the stomachs of Goosanders shot in the lower Tweed area.

NUMBER OF LAMPREYS		DATE	ORIGIN	Total Birds	With Food
River Lamprey	1	1991 Apr	Coldstream - Estuary	31	29
River Lamprey	4	1992 Feb	Coldstream - Estuary	15	13
River Lamprey	1	1992 Aug	Melrose - Estuary	3	3
River Lamprey	17	1993 Mar	Coldstream - Estuary	23	23
River Lamprey	3	1993 Apr	Coldstream - Estuary	10	10

These birds were shot during licenced culls made during the salmon smolt migration period, February to April, which coincides with the end of the upstream migration period for River Lamprey and the beginning of their spawning. The seasonal distribution of River Lamprey in the diet of these birds does, in fact, show their occurrence peaking in April. Table 1.3 below is for the whole Tweed catchment, not just the lower as in Table 1.2:-

Table 1.3 Seasonal occurrence of adult lampreys in Goosander stomachs collected frm the Tweed catchment

Number of Stomachs	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	TOTAL
Total	6	7	9	2	1	0	1	0	0	2	1	29
With River Lamprey	1	3	6	0	0	0	1	0	0	1	0	12
With Brook Lamprey	1	1	2	0	0	0	0	0	0	0	0	4

Surveys of lamprey larvae in and around the Till catchment

1.11 No Sea Lamprey larvae have been found in the Till in any of the juvenile surveys made so far (Johns, 1996; Campbell 2002; Campbell, 2013) though they have been found in the Tweed both upstream and downstream of the Till confluence (Campbell 2004 and Campbell 2006)

1.12 River Lamprey larvae were identified in the Till and its upper portion, the R. Breamish by Johns (1996) as shown in Table E1 in Appendix E taken from his report. However, it is not clear on what parameters the identifications as River Lamprey were made. The note to his Site 9 is "*River Lamprey ammocoetes and newly emerged adults were found in this clay. Adults were no more than 12cm long …. River and brook Lamprey present here (<10cm long). All had no eye development*." The note to Site 10 has "*Many River Lamprey ammocoetes of 12cm length that have eye development*". The note on Site 12 is "…. *Many brook and river Lamprey ammocoetes*". This report was made before the *Natura 2000* publications on lamprey had been produced and it is now accepted that River and Brook Lamprey ammocoetes are indistinguishable in the field. It is not therefore safe to accept the ammocoete identifications made in this report, though eyed individuals of 120mm or less are very likely to have been River Lamprey, as shown below.

1.13 Gardiner (1995) gives the size range of Brook Lamprey at the time of metamorphosis as 130 to 150mm and that of River Lamprey as 90 to 120mm. A population of ammocoete larvae without any individuals over 120mm is therefore likely to be of River Lamprey and two such populations were found in the 2002 survey, at Twizel and at Tindall, both on the lower river (Table E.2, Appendix E). The 2013 survey of the Till also found no ammocoetes larger than 120mm at these same Twizel and Tindall sites (Table E.3, Appendix E). That two surveys 11 years apart, the 2002 at the end of July and the 2013 in the second week of September, should not have found any individuals over 120mm out of the 216 (Twizel) and 156 (Tindall) sampled is highly suggestive of River Lamprey spawning, an impression re-inforced by the 90mm long macropthalmia sampled at Twizel in 2013.

1.14 The Twizel and Tindall sites are near the bottom of the Till, and so in easy reach of migratory lamprey, but the 2002 and 2013 surveys also suggest that there might be River Lamprey spawning further up the river. In 2002, the site at Redscar produced only two larvae greater than 120mm out of 76 sampled and macropthalmia of 110, 115 and 115mm (Table E.2, Appendix E) and in 2013, only two of over 120mm out of 30 sampled and a macropthalmia of 115 (Table E.2, Appendix E). Much further upstream, about three-quarters of the way up the Till, the site at New Bewick produced only one larvae larger than 120mm of the 153 sampled in 2002 as well as a 95mm macropthalmia (Table E.2, Appendix E) and in 2013, only three out of 70 larvae at this same site were over 120mm with macropthalmia of 100 and 110mm (Table E.3, Appendix E). Again, while not conclusive, these larval and macropthalmia sizes are very suggestive of River Lamprey spawning.

1.15 As said above, while differences in larval length–frequency distributions are possible distinguishers of Brook and River lamprey populations with the former being generally older and larger at metamorphosis this is not an absolute guide. Kelly and King (2001) report that the River Lamprey has a "*relatively short*" larval life of 4.5 years, compared to an estimated 6.5 for Brook Lamprey but add that "*there are large differences in growth rate because of local variation, nutritional factors, stability of larval beds and the degree to which the population is subjected to washing-out*". They also quote Malmqvist (1978) as finding that the duration of larval life varies within populations and summarise the topic by saying that metamorphosis takes place "*after a period of 3-8 years in freshwater, depending on species and location*".

2 METHODS

Spawning habitat surveys

2.1 The surveys to locate areas of spawning habitat were made on foot and upstream along roughly equal length sections of the channels, as listed in Table 2.1:

Table 2.1 Survey sections and dates they were walked

Channel	Section Number	From:	To:	Date : April
TILL	T 01	Tillmouth	Ford ford	6th
	T 02	Ford ford	Redscar Br.	7th
	T 03	Redscar Br.	Doddington Br.	7th
	T 04	Doddington Br.	Lyham Br.	9th
	T 05	Lyham Br.	Newtown Br.	10th
	T 06	Newtown Br.	New Bewick Br.	10th
	T 07	New Bewick Br.	Powburn Br.	13th
BREAMISH	B 01	Powburn Br.	Ingram Br.	13th
	B 02	Ingram Br.	Linhope	13th
GLEN / BOWMONT	G 01	Till confluence	Lanton Weir	14th
	G 02	Lanton Weir	Kilham Br.	14th
	G 03	Kilham Br.	The Border	15th
WOOLER W /	W 01	Till confluence	Cauldgate Mill	17th
HARTHOPE B	W 02	Cauldgate Mill	Howsen Burn	20th
LILBURN	L 01	Till confluence	Middleton track	20th

The foot surveys were predicated on the peculiar nature of the Till, which has long sections of sandy channel unusable by any gravel-spawning fish species. This restricts available spawning and the prime purpose of the survey was to map where these sandy sections were and so define where spawning could be possible and so where further surveys and assessments of lamprey spawning should be targeted. However, there also turned out to be long stretches of flat, bedrock strata in the lower sections which must further restrict spawning opportunity.

2.2 The data recorded during the foot surveys were therefore of the basic types of channel & substrate seen, referred to as “Zones”. These were:-

a) "**Natural Sequence**" : the typical sequence of Riffles / Runs / Glides / Pools seen in rivers, with a note made as to the size of substrate as in some sections of the Till this sequence was apparent but with only Pebbles and / or smaller substrate types present.

b) "**Slow : Bedrock**": in the lower Till, long stretches of this are present. The bedrock is flat and smooth and in strata, with rock sills where the bed of the river drops from one stratum to the next. There can be deep holes in these layers however, but not filled with Gravel or Cobble as might be expected. This zone type could also be either **Deep** or **Shallow.**

c) "**Slow : Sand / Silt**": This type predominates on the middle Till and some of the sections were wholly of this form. Variants on this type were "**Slow: Sand / Silt Glides with Pebble Riffles**"; "**Slow: Shingle**" and "**Slow: Gravel**"

Minor variations on these types were recorded where these occurred. There were also short stretches of a different habitat type that were recorded as such, but were regarded as being too short to be counted as zones e.g. the occasional large pool. A new zone was started upstream of such "singularities" however.

2.3 Maitland (2003) described the spawning habitat of River Lamprey as *"… where there is some water flow through the substrate. This is usually found at the tails of pools where the gravels have been deposited from upstream and the scouring of pools but the current is still reasonably fast. This is broadly similar to the situation favoured by stream salmonids and is the area where there is maximum penetration of gravels by water currents. The particle size is variable, but is usually described as gravel with some sand …*." He also quoted reports that Sea Lamprey use gravel of 10-120mm in diameter, with some sand present.

2.4 Substrate suitable for lamprey spawning was recorded on the maps where it was exceptional e.g. in areas of Natural Sequence where there was a mix of substrates from Cobble to Sand, lamprey could spawn in many places, so it was only of locations where site and substrate were exceptionally suitable that a record was made within such zones. Where the dominant zone type was **Slow: Sand / Silt** or **Slow: Bedrock**, suitable spawning was rarer and was recorded on the maps wherever it was found as one of three grades:

SP1 Pebbles, Gravel & Sand at the tail of a pool or a glide. (>10 m2 area).

SP2 Pebbles, Gravel & Sand at sides in flowing water (<10m2) or <10m2 of SP1

SP3 Pebbles, Gravel & Sand mixed with larger substrates, at sides or at tail of pool or glide,

or SP3 ii) a limited area of SP2 habitat at sides (>10m2) or

SP3 iii) absence of Pebble substrate but presence of gravel & sand in fast flowing water.

In some situations, more suitable habitat for spawning was artificial, due to materials brought in for bank repairs or other human activities within zones that were otherwise largely devoid of usable substrates. Data was recorded on large scale maps and on survey sheets, from which maps were constructed.

Instream structures survey

2.5 During the habitat surveys, notes were made of instream structures – caulds (large weirs), weirs, fords and bridge aprons – that could be obstructions, either total or partial. A number of the mill dams marked on the maps now no longer exist: Heiton Mill, Barleymill, and Etal and two fords marked on the map have no structure: Coupland (R. Glen) and Brandon (R. Breamish). Those structures listed as possible obstructions to River or Sea Lamprey passage were later re-visited for measurements to be taken as a basis for assessment.

The measurements taken were based on Annex 6 of the assessment criteria for River and Sea lamprey passage being worked out for the Water Framework Directive (currently available at:

http://www.wfduk.org/sites/default/files/Media/Environmental%20standards/Annex%206%20Rivers%20Fish%20FCS2%20%26%20Fish%20Barrier.pdf

and given in Table 2.2

Table 2.2 Criteria for adult migratory lamprey passage of instream structures, taken from Annex 6 – RIVERS – Fish – FCS2 and Fish Barrier Tool.

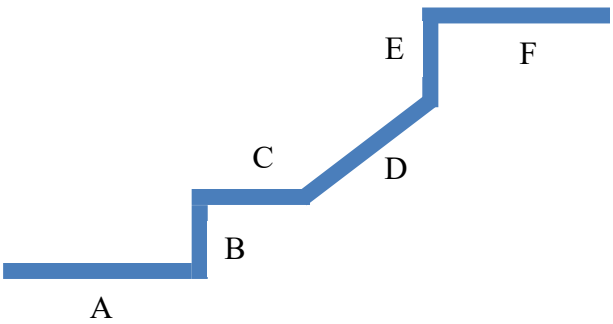
Barrier type	Criteria	Adult lamprey (sea/river)
Obstacle presenting a vertical drop	Vertical hydraulic head	≥0.3 metres

Obstacle presenting a slope (including culverts)	% Slope of structure	
	o Structure effective length ≤ 3m	≥ 40%
	o Structures effective length 4-9m	≥ 20%
	o Structures effective length ≥ 10m	≥ 15%
Effective length of obstacle (the distance which would have to swum by a fish to pass the barrier)		≥50 metres

Table 1: Critical physical limits for barriers to the main species. If dimensions of a barrier are greater than any of these limits, it should be considered to be a "severe impairment" to the species.

The values in this table are taken from the relevant tables in the WFD111 project manual

2.6 As would be expected, the various different instream structures were of very varying shape and form and were seldom a simple slope or single vertical face as listed in Table 2.2. The measurements taken were therefore defined as one of the structural components shown in Figure 2.1



A	Flat below vertical
B	Vertical
C	Flat above vertical
D	Slope
E	Vertical above slope

F	Flat above slope

Figure 2.1 Possible components of instream structures measured

Not all these features were present at all the structures, but defining them in this way allowed the measurements taken at the structures to be tabulated together for comparison.

Six structures had associated fish passes, and one had both a fish and an Eel pass and these were also measured for assessment for lamprey passage. Photographs were taken at all structures and fish passes.

Redd / Spawning Surveys

2.7 Based on the habitat surveys, sites and areas of suitable habitat (para 2.3 above) for spawning that were also of easy access were identified, the access criterion being so that repeated visits could be easily made as redds can become rapidly obscured. The sites defined and visited are given in Table 2.3

Table 2.3 Sites inspected for signs of spawning activity

AREAS SURVEYED R. TILL	Habitat Section	Ref no	SURVEY SITE GPS
Tillmouth to Twizel Cauld	T01	01	NT 87014 43012 (in Tweed)
	T01	02	NT 88177 43209
	T01	03	NT 88302 93303
	T01	4a	NT 88613 43018
	T01	4b	NT 88654 42844
	T01	4c	NT 88686 42700
Upstream of Ford Bridge	T02	01	NT 93847 37485
	T02	02	NT 93751 37337
	T02	03	NT 93582 37212
	T02	04	NT 93433 37086
Upstream of Doddington Br	T04	01	NU 00013 30720
	T04	02	NU 00115 30726
	T04	03	NU 00207 30707
	T04	04	NU 00246 30703
	T04	05	NU 00221 30643

	T07	01	NU 07658 20338
Upstream of New	T07	02	NU 07802 20210
Bewick Br	T07	03	NU 07804 20039
	T07	04	NU 07882 19907
(Harehope Br.)	T07	05	NU 07901 19867
	T07	06	NU 07870 19774
WOOLER WATER			
	W01A	01	
Glide ends	W01A	02	NT 99519 28615
	W01A	03	NT 99564 28698
	W01A	04	NT 99838 29581
Between fish passes	W01B	01	
R. GLEN			
Coupland Ford	GO1	01	

KNOWN SPAWNING SITES OF LAMPREY IN THE MAIN TWEED
Mertoun Bridge (middle Tweed)
Coldstream Bridge (lower Tweed)

The first of these sites, T01 (Tweed gravel banks) is actually off the mouth of the Till where it meets the Tweed, but was included as a likely place for spawning. Two known sites for Sea and River Lamprey spawning outwith the Till were also checked as reference points, upstream of Mertoun Bridge on the middle Tweed, and downstream of Coldstream bridge on the lower Tweed. Although migration is largely nocturnal, at spawning time the nocturnal preference is lost and spawning activity is as frequent in daytime as at night (Jang & Lucas, 2005)

2.8 The key sites for both River and Sea Lamprey spawning were thought to be those downstream of Twizel Cauld, so these were checked most frequently. Instream structures that can hinder migration can cause accumulations of spawning lamprey (Lucas & others, 2009) and the Twizel area is the first with significant amounts of gravel that would be encountered by any lamprey migrating upstream – the study of Jang & Lucas (2005) was centred on the first significant area of gravel that would be encountered by River Lamprey migrating up the Yorkshire Derwent. While the Twizel area fitted both these parameters, there are, however, no known reports of spawning lamprey there

2.9 Redd surveys depend on good weather and water conditions, especially in larger channels. Redds are identifiable as "*depressions in the gravel surrounded, at the rear and sides, by gravel moved outside the depression ….also characterised by clean gravel and sand from the overturned and disturbed material, compared to the surrounding gravel that was covered with epilithic algae*" (Jang & Lucas, 2005). Spates, however, remove the signs of the physical disturbance and algae will re-grow in a matter of days and visibility can be reduced or eliminated by coloured water and wind action on the surface. In the five year study of Lucas & others (2009) on the Yorkshire Derwent, redd surveys were not possible in two of the years due to high flows at spawning time.

3 RESULTS

HABITAT SURVEY

3.1 The numbers and types of habitat "zones" found during the walking surveys are listed in Table 3.1. Even in this very basic format, the main features of the Till catchment are apparent: the bedrock areas of the lower Till; the slow flowing sandy areas of the middle Till and lower Glen and the natural sequences of Riffle – Pool and Cobble-Gravel in the upper Till and in the tributaries. The distribution of these habitat types is mapped in Figure 3.1 and the detailed survey information is given in Appendix 1 (draft). Where it seemed practicable, a spawning grade was given to a whole habitat zone as well as to specific spawning sites, but it was not always possible to do this given the complexity of some zones.

Table 3.1 The numbers and types of habitat zones found

River & survey section	Ref No.	Number of habitat zones recorded				Percentage occurrence		
		Natural Sequence	Slow & Bedrock	Slow & Sand	Total	Natural Sequence	Slow & Bedrock	Slow & Sand
TILL	T 01	6	12	1	19	32%	63%	5%
	T 02	1	1	4	6	17%	17%	67%
	T 03	1		1	2	50%	0%	50%
	T 04	1	2	11	14	7%	14%	79%
	T 05			1	1	0%	0%	100%
	T 06	2		1	3	67%	0%	33%
	T 07	3		2	5	60%	0%	40%
BREAMISH	B 01	1			1	100%	0%	0%
	B 02	1			1	100%	0%	0%
GLEN / BOWMONT	G 01	1		3	4	25%	0%	75%
	G 02	1		1	2	50%	0%	50%
	G 03	1			1	100%	0%	0%
WOOLER W / HARTHOPE B.	W 01	3			3	100%	0%	0%
	W 02	1			1	100%	0%	0%
LILBURN	L 01	2			2	100%	0%	0%

3.2 The survey section with the least spawning was T03, Redscar Bridge to Doddington Bridgen where in 9.6kms of river channel, only four spawning sites, all of the lowest, SP3, grade were found (Appendix 1) in amongst the dominating sand substrate. This does show however, that even in the sandiest zone, there were a few possible spawning sites. The best sections for spawning were probably B01, the R. Breamish from Powburn up to Bulbies Wood and G02, the middle Glen, from Coupland Ford up to Kilham Bridge.

3.3 The overall habitat map (Figure 3.1) shows how the best spawning areas are in the upper parts of the main Till and the middle Glen and in the smaller tributaries, the Wooler Water and the Lilburn.

4 Discussion

The suitability of the Till for migratory lamprey spawning

4.1 As is clear from the habitat maps (Figs. 3.1) and the data in Appendix A, the Till has plenty of suitable habitat for both spawning and larvae, but these are not much intermixed. The larval habitat is concentrated in the main channel in the flatter, lowland, zone while the spawning habitat is in the tributaries where the gradients are steeper. While this is unusual for the Tweed catchment it is similar to the situation describe for the Yorkshire Derwent by Lucas & others (2009):-

"*Silt and sand (habitat for lamprey larvae) is abundant throughout the lower river, but likely spawning habitat is most common in the middle and upper reaches of the river and in localised areas below the weirs. The Derwent's main tributary, the Rye, joins at rkm68. Swift shallow water with gravel / cobble beds become progressively more abundant in the Derwent, Rye and minor tributaries upstream of the Rye-Derwent confluence…… the main areas of gravel spawning habitat at low to intermediate gradients occur in the Rye and upper Derwent subcatchments. Within the lower 68 km of the Derwent main channel, following intensive searches, a total of 0.4% river length was ascribed as being potentially suitable for River Lamprey spawning. Spawning habitat was much more abundant upstream of that area, associated with elevated gradient and related changes in stream geomorphology, especially in the main tributary, the Rye. Despite underwater camera and snorkelling observations, no gravel spawning habitat was found in deep water areas."*

This Derwent study radio-tracked River Lamprey upstream in three consecutive years, the median distances travelled being 34, 27 and 35 kms, the maximum distances in all three years being 45kms. It appears therefore than none of the lamprey tracked in this study would have made it as far upstream as the beginning of the zone of more abundant spawning which began at rkm68, with the confluence of the R. Rye. Whether any would have reached this zone if it had not been for the impact of the weirs downstream (64% made it past the first, only 17% past the second) is uncertain.

4.2 For the Till, however, there are two local pointers as to how far River Lamprey can migrate upstream. The first is the capture of three specimens in a smolt trap at spawning time in 1999 and 2000 on the Leader Water, at Drygrange, which is 70kms (altitude 90m) from salt water (taken as the Bypass Bridge at Berwick) and the second is the finding of River Lamprey during the fish rescue at Kirknewton on the R. Glen in 2008, which is about 55kms (alt. 61m) from salt water. Again, it should be noted that the Leader individuals had passed seven instream structures on the main Tweed to reach their capture location. There are actually historic references to River Lamprey (probably) going even further upstream in the Tweed catchment than the Leader Water. Sir Thomas Dick Lauder, reminiscing in 1848, wrote that at the Bridge of Yarrow : "*It is remarkable that lampreys used to come up here to spawn, and if we do not mistake, we ourselves saw several sticking to the stones, near the bridge, like floating pieces of tangle*". (Lauder, 1874). This more likely to be a reference to River Lamprey given the comparison with "tangle" (Kelp, *Laminaria* spp) as the stems of this are really too thick to be compared with Brook Lamprey. There is also a memory of River

Lamprey spawning at Lindean, near the foot of the Ettrick W in the 1950s (D. Mitchell, *pers. comm*.) and the Yarrow W. is a tributary of the Ettrick. Lindean, on the Ettrick, is a further 11.5kms upriver from the mouth of the Leader and the first bridge on the Yarrow W. is 19.5kms upstream of the Leader (it is not clear which bridge is meant by "bridge of Yarrow".

4.3 As shown on Figure **, the more upland, coarser-substrate zone of the main channel of the Till / Breamish begins at ** which is ** from the sea. The Wooler Water joins the Till at 50kms (alt. 38m) from the sea and the Lilburn at 73kms (alt. 54m), so the first is certainly within reach and the second probably is. A distance of 70kms from the sea up the Glen / Bowmont would take a River Lamprey as far as its headwaters around Cocklawfoot.

4.4 There are therefore no intrinsic reasons of distance or altitude as to why most of the Till catchment should not be usable by River Lamprey for spawning. However, there are no records of the sort of numbers of Lamprey seen spawning in, for example, the Humber catchment where there is a commercial lamprey fishery on the tidal R. Ouse that catches 20-40,000 River Lamprey annually (Jang & Lucas, 2005). There are no records of lamprey of any sort being commercially fished on the R. Tweed.

History of instream structures as potential obstacles to lamprey migration

4.5 There is a considerable history of barriers on the Till e.g. Francis (1874) referred to the cauld at Ford (Heatherslaw) saying it "*sadly needs to be done away with or greatly mitigated*" and to the then cauld at Coupland, on the R. Glen which he regarded as being a barrier that Sea-trout could only get over in exceptional flows. He also referred to a weir at Lanton (presumably the base for the E.A. gauging weir now there) as being less difficult for Sea-trout than the Coupland one downstream of it. On the 1899 O.S. map there are another two weirs on the Glen, at Canno Mill and at Mindrum, which may have been later than his time. Surprisingly, he does not refer to the cauld at Twizel Mill at the very foot of the Till, though a witness to a Commission of Inquiry in to the Operation of the Tweed Fisheries Acts (Walpole & Young, 1874) complained of the way that netsmen took the fish held up by this. Calderwood (1909) referred to four weirs in the Till catchment interestingly saying "*the first, at Twizel, being of little moment*" but that the then cauld at Heaton Mill, referred to as "*not now worked*" was a difficulty for fish due to the "*heavy rush of water*". The cauld at Etal is referred to as being even more of an obstruction and that it was reported that few salmon were taken in the upper Till because of this structure. He also, however, states that Sea-trout found less difficulty in passage and penetrated to the R. Glen / Bowmont and to the College Burn, its main tributary. In all, some 10 caulds & weirs, past and present, have been identified in the Till catchment by the Tweed Foundation.

4.6 Systematic work on improving access seems to have started in the 1960s, judging by an item in the Annual Report for 1967 of the River Tweed Commissioners:

"A fish pass was designed by the Commissioners staff and installed by the Northumberland County Council at the Hedgeley Road Bridge cauld on the Breamish. Not only were a number of Salmon seen to use the pass, but a number of dead salmon kelts were found in the area. No salmon and very few sea-trout have previously been seen at this point. The Till and the Breamish dams all having been made passable, the Commissioners are of the opinion that the result justifies their policy in relation to caulds."

The 1968 Annual Report specified that the caulds at Etal and at Ford Forge (now called Heatherslaw Mill cauld) had had "*partial slaps*" made in 1964 that had now been completed to the Commissioners satisfaction – the Superintendent of the time having the view that slaps would be more effective than fish passes. A "*slap*" is the technical term for (1) a gap in a weir or cauld made to allow fish passage and (2) a period of time in which no net fishing is

allowed e.g. the "*weekly slap*" on Saturdays and Sundays on the Tweed. It could also refer to a period in which mills were not allowed to operate so that all the water they used remained in the river to assist fish passage. The minutes of the RTC's Police Committee for the 7th September 1964 also recorded that slaps were made in the caulds at Twizel Mill and at Heaton Mill. Before the slap was made in the Twizel cauld, there was a length of it about four yards long that was lower than the rest of the crest and this served as the fish pass (Michael Chapman, *pers. comm.*). When the present fish pass was built in 1987 (modified in 1988), the slap was closed up (RTC files).

4.7 All these observations and works refer to the passage of Salmon and Sea-trout, so it is not easy to interpret what they would have meant for Lamprey. However, two personal observations give some information on this:

(1) From Michael Chapman, at one time a netsman at Twizel Mill and for many years the head boatman of the Tillmouth fishings: He remembers about 30 years ago being with a John Swan who tried to net the fish they saw trying to make it through the then slap in Twizel Cauld. However, these slipped through his landing net because, it turned out, they were Lamprey (as this was illegal, this was at night, when River Lamprey run).

(2) From Julia Nolan, manager of Heatherslaw Corn Mill: Remembers being told that at least 20 years ago, in her predecessor's time, Eel and Lamprey used the mill lade to get upstream, so that the "*water looked as if it was going in reverse*".

These are the only recollections obtained so far that refer to mass numbers of lamprey being visible in the Till catchment. Both relate either to the period when there was a full sized slap in Twizel cauld or not long afterwards. It may be that access through this was easier than through the fish pass that was subsequently built, but this could not be an explanation for any diminution of a previously substantial River Lamprey population as before the slap was made the cauld was intact, with only a lowered length of crest provided for fish passage. Judging by modern studies of lamprey passage at weirs (Jang & Lucas, 2005 and Lucas & others, 2009) Twizel cauld before the slap was made would have been a partial barrier to lamprey. That it was not a total barrier to fish that cannot jump is shown by the references to Eel in the Till catchment made in the fishing memoirs of Henderson (1879) which come from the latter 19th century. The earliest reference known to a cauld at Twizel is 1874 (in Walpole & Young, 1874)

4.9 The map of ammocoete distribution in the Till catchment (Figure E.1, Appendix E) shows them to be widespread downstream of major natural obstacles, indicating that at some period River Lamprey spread colonised the catchment sufficiently to produce resident Brook Lamprey populations. However, this could have been at any time between the end of the Ice Age and the construction of the first weirs. Certainly, by the time that natural history observations for the Till catchment become available in books such as "*My Life as a Angler*" (Henderson, 1879) in which many notices of wildlife can be found, there are none of lamprey. On the other hand, the annual "Histories" of the Berwickshire Naturalists Club have not yet been examined for information on lamprey.

4.10 On other rivers where lamprey are blocked their numbers spawning below barriers can be very visible (e.g. Lucas & others, 2009), but Michael Chapman, the head boatman of Tillmouth fishings (which included the lower Till till recently) has never seen this either at Twizel Mill cauld or at any of the ruined weirs between it and the mouth of the Till.

The present day potential for spawning of migratory lamprey in the Till catchment and answers to the study questions.

4.11 As shown above, there appear to be no intrinsic reasons of distance from the sea; altitude or lack of spawning or larval habitat as to why the Till catchment should not be widely used by spawning migratory lamprey. A summary of the assessments of the passability of significant instream structures made during this study is given in Table 4.1

Table 4.1 Passability assessments of instream structures in the Till catchment to Lamprey

Ref	Location	Structure	Fish pass	Eel pass [(a)]	Eels upstream?
E	Twizel cauld	**X**	x		Yes
K	Powburn Bridge	**X**	x	?	No
N	Lanton Weir	x	none		Yes
O	Wooler Cauld	**X**	x		Yes
P	Wooler Br. apron	**X**	O		Yes
Q	Haughead metal piling	**X**	none		Single record only
R	Haughead ford	**X**	O		Single record only
V	Lilburn gravel trap	x	O		Yes
W	Lilburn ford	x	none		Yes
X	Lilburn bridge apron	O	O		Yes

KEY

(a) Only at Powburn

Impassable at all flows	**X**
Impassable at low flows	x
Passable at all flows	O

4.12 The assessment is that there are two serious blocks to lamprey migration within the Till catchment:-

a) the iron pilings at Haughead on the Wooler Water (Plate C.13, Appendix C). However, the water held up by these make the fish pass at the ford just upstream accessible at all flows. If it was not for these pilings, this would be inaccessible at low flows. There are, effectively, no Eels found upstream of this structure, showing its effect on fish that cannot jump.

and

b) the bridge apron at Powburn. This is a massive structure with both fish and eel passes. The fish pass is probably unusable by lamprey at low flows and the Eel pass is marginal judging by its length and slope, although its “brush” lining may assist lamprey. At the time of inspection, though, the eel pass was completely blocked with gravel and, judging by the size of some small trees growing the gravel blocking its entrance, had been so for two to three years. As the pass was constructed in 2011, this suggests it has been blocked for much of its existence. No Eel have been found upstream of Powburn since 2007, showing that the previous fish pass was not usable by fish that cannot jump and that the new passes, fish and Eel have not, as yet, made any difference to this situation. (Plates C.4-8, Appendix C)

At present, the Tweed Foundation is electric-fishing upstream of both these structures annually, in the case of Haughead, to collect evidence on how it is blocking salmon migration (Sea-trout do not appear to have much problem with it) and for Powburn, to monitor how the population of juvenile salmon upstream is increasing since the new pass was made.

4.13 The other significant instream structures, other than Lilburn ford, all have fish passes, which are judged either to be usable at all flows or unusable only at low flows, which are not likely to inhibit winter migrants such as lamprey. At high flows some of the structures themselves will be passable and Lucas & others (2009) found that flows higher than Q10 can be utilised by some River Lamprey to get over substantial barriers.

4.14 There are large areas of suitable spawning substrate, although these are largely upstream of the main areas of substrate suitable for larvae. This “separation” of the two habitat types in to upland and lowland zones should not, however, cause a problem to migrating lamprey as local Tweed and Till records show River Lamprey able to travel the necessary distances and altitudes to reach these areas of spawning habitat.

4.15 Indications from juvenile surveys suggest that River Lamprey do use the lowermost Till for spawning, and that there could be some spawning further upstream. There are, however, no recent records of River Lamprey running the river, or spawning, in numbers that would be noticeable by casual observers. The lack of any sightings of spawning activity during this study suggests that numbers utilising the Till catchment are low.

4.16 The widespread present day distribution of ammocoetes does suggest that River Lamprey at one time penetrated the Till catchment widely, to produce local populations of Brook Lamprey but the history of obstacles to salmon and sea-trout migration indicate that lamprey access would have started to be interrupted many years ago.

4.17 No sightings of Sea Lamprey were made though one record was found of what was very probably spawning at Tindal House about 15 years ago.

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Figure 3.1 Map of the different habitat types found (*draft map*)

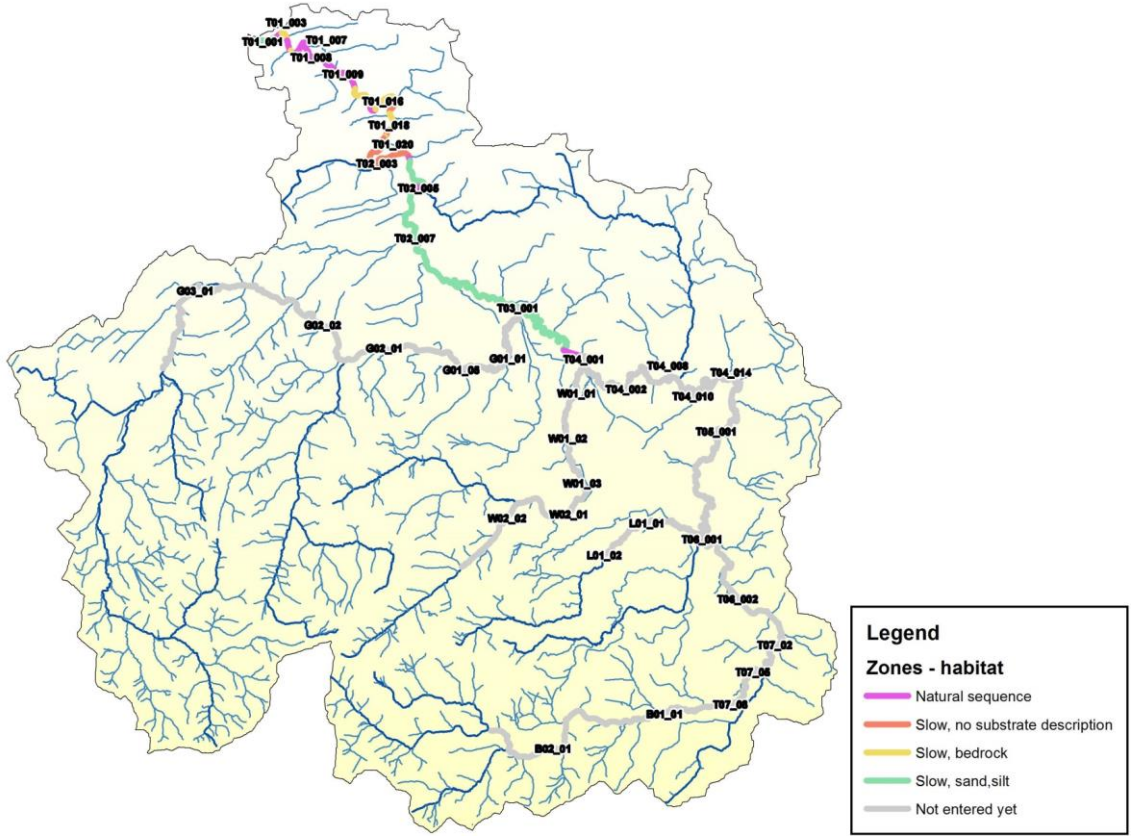


Figure 3.2 : Habitat zones from the confluence of the Tweed to the Wooler Water

Figure 3.3 Habitat zones from the confluence of the Wooler Water to Linhope on the R. Breamish

Figure 3.4 Habitat zones on the R. Glen

Figure 3.5 Habitat zones on the Wooler Water

Figure 3.6 Habitat zones on the Lilburn

Figure 3.7 Channel profile of the R. Till

Figure 3.8 Channel profile of the R. Glen

Figure 3.9 Channel profile of the Wooler Water

Figure 3.10 Channel profile of the Lilburn

OBSTACLES SURVEY

3.4 The aim of this survey was to record all instream structures which could limit migration taking account of both low and high flow conditions and also identify those that would be complete migration barriers in any flow.

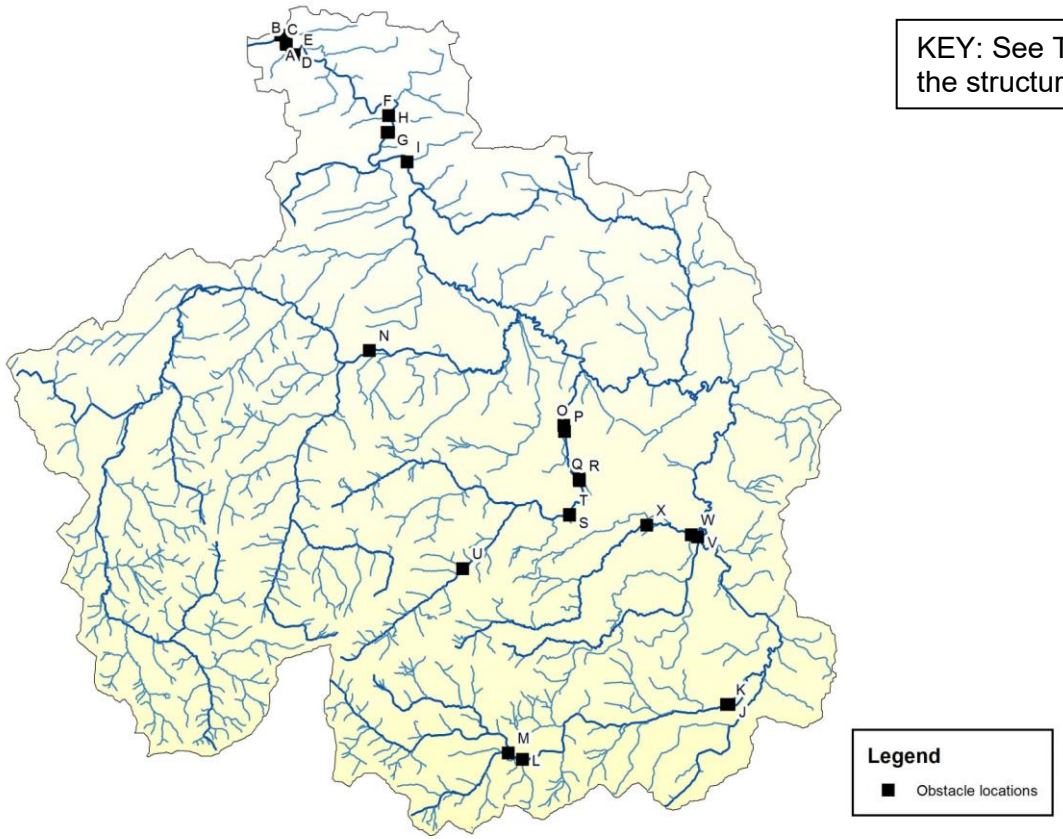
3.5 The list of structures significant enough to be potential barriers that were found during the habitat surveys is given in Table 3.2. Minor structures or those obviously passable at a wide range of flows, such as the apron under Weetwood bridge (NU 029 296) a ford near Harehope (NU 077 195) or the Coldgate Mill ford on the Wooler Water (NT 998 248) have been omitted. This table also shows which structures are downstream of the only record apparently ever made in the Till catchment of adult River Lamprey (Plate 1.1).

3.6 Table 3.2 also gives the available data on the distribution of ammocoete larvae and Eel in relation to the listed structures as an additional guide to assessing their passability to lamprey (Tweed Foundation electric-fishing data). The significance of the presence of Eel upstream of a structure is that it shows it can be passed without jumping. Ammocoete populations can be those of resident Brook Lamprey populations but do show that there is suitable habitat for juvenile lamprey upstream of structures. Eels can be up to 20 years old in this part of the world, so their simple presence upstream of a structure need not necessarily indicate recent passability, though that of small, young, Eels would. The timed electric-fishing data collected by the Tweed Foundation does not, however, include size of Eel.

3.7 The assessments made of those structures that were considered to have potential for affecting lamprey passage are listed in Table 3.3, the parameters for assessment being those listed in Table 2.2. The assessments made in this table are for the physical structures themselves, not taking account of any fish passes which are assessed separately in Tables 3.4 and 3.5. The particular difficulty here is in assessing the problems that the vertical heights of obstacles (or of fish pass partitions) might cause. At low flows a higher amount of such verticals is exposed but as the flows increase and downstream water levels rise, this becomes less and less and the vertical may become entirely flooded over. The question then arises as to whether fish would be moving at such high flows, which will also be related to water temperature.

3.8 Assessing structures for passability from their appearance at one time of year and extrapolating this to how their flows and forms interact to affect lampreys during the months of their migration over the winter and spring can only give a general indication of passability and, of course, even if some fish pass, it does not mean that all that wanted to pass were able to do so. However, some structures are clearly not passable under any conditions by fish that cannot jump, such as the vertical sheet metal pilings downstream of the Haughhead ford and fish pass or by any fish at all, such as the Hedgeley Bridge apron. For some structures, there can be ways round at the channel sides at high flows and Lucas & others, (2009) found that radio-tracked River Lamprey in the R. Derwent (Yorkshire) could utilise flows very much higher than those they migrated on to get over structures, even flows less frequent than Q90. Ability to utilise such high flows makes prediction of the effect of structures on lamprey migration very difficult, though it is probably safe to assume that even if some manage to pass on very high flows, others will fail.

Figure 3.2 Locations of significant instream structures



KEY: See Table 3.2 for the names of the structures indicated by the letters.

Table 3.2 Significant instream structures found on the Till and Tributaries (Photographs in Appendix 02)

HABITAT SECTION		STRUCTURES & GPS WATERFALLS	Condition	Evidence for adult lamprey passage	Upstream Records [a] Ammocoetes	Eel
		R. TILL				
T01	A	Tillmouth 1st weir	Broken		+	+
	B	Tillmouth 2nd weir	Broken		+	+
	C	Tillmouth 3rd weir	Broken	*River Lamprey adults found upstream of these during fish rescue on the R. Glen at Kirknewton 2008-11-03*	+	+
	D	Twizel weir	Broken		+	+
	E	Twizel cauld & fish pass	Intact		+	+
	F	Rock sill with fish pass	Has breaks		+	+
T02	G	Etal Cauld	Broken			
	H	Etal ford	Intact		+	+
	I	*Heatherslaw Mill Cauld*	*Natural slope left*		+	+
T07	J	*Hedgely railway br. apron*	*Undercut*		+	+
	K	Powburn bridge apron & fish & eel passes	Intact		+	-
		R. BREAMISH				
B02	L	Waterfall	Natural		-	-
	M	Waterfall	Natural		-	-
		R. GLEN				
G02	N	Lanton gauging weir	Intact	*R. Lamprey adults upstr. 2008-11-03*	+	+

HABITAT SECTION		STRUCTURES & GPS WATERFALLS	Condition	Evidence for adult lamprey passage	Upstream Records [a]	
					Ammocoetes	Eel
		WOOLER WATER				
W01	O	Wooler cauld & fish pass	Intact		+	+
	P	Wooler br. apron & fish pass	Intact		-	+ [b]
	Q	Metal piling downstream of Haughhead ford	Intact		-	+ [b]
	R	Haughhead ford & fish pass*	Intact		-	+ [b]
	S	Series of six weirs	Intact		-	+ [b]
W02	T	Waterfall	Natural		-	-
	U	Waterfalls	Natural		-	-
		LILBURN				
L01	V	Lilburn gravel trap & fish pass	Intact		+	+
	W	Lilburn ford	Intact, side passage		+	+
	X	Lilburn bridge apron & fish pass	Intact		+	+

(a) Combined records from electric-fishing surveys made in 2007, 2010 & 2013

(b) Only at one site out of eight in all three survey years. Haughead ford was passable in the 1990s when an artificial rapids was in place downstream of it, making a slope up to it and the fish pass then in place.

* Scheduled to be "made good" in the next WFD cycle

Table 3.3 Assessment of passability of significant instream structures for lamprey (Photographs in Appendix 02)

		Structure components (see diagram): All lengths and heights are in metres							
INTACT STRUCTURES	Total height [(a)]	Flat below vertical	Vertical	Flat above vertical / below slope	% Slope	Effective length of slope	Vertical above slope	Flat above slope	Route around side(s)
Position on diagram **		A	B	C	D		E	F	
FP denotes fish pass and EP Eel pass but measurements are for main structure only									
R. TILL									
A: Twizel cauld (FP) (part vertical, part slope)	1.5	-	1.5	-	40%	4	0.27	0.8	No
H: Etal ford	0.3	-	0.3 [(b)]	5.0	-	-	-	-	Yes
K: Powburn bridge apron (FP) (EP)	2.7	-	2.7	-	-	-	-	-	No
R. GLEN									
N: Lanton gauging weir	0.75	-	0.6	2.6	11.1%	1.4	-	-	No
WOOLER W.									
O: Wooler cauld (FP)	2.0	-	0.4 <> 0.65	2.55	36.2%	4.45	-	0	No
P: Wooler bridge apron (FP)	1.5 to 1.8	-	0.6 <>0.9	3.25	30.5%	3.4	0.05 to 0.15	19.0	Yes
Q: Metal piling downstream of Haughead	1.6	-	1.6	-	-	-	-	-	No
R: Haughhead ford (FP)	2.5	-	2.4	9.6	-	-	-	-	No
S: *The series of six weirs were not assessed as they are not reachable due to the major obstructions downstream*									

Table continued

		Structure components (see diagram): All lengths and heights are in metres							
INTACT STRUCTURES	Total height [(a)]	Flat below vertical	Vertical	Flat above vertical / below slope	% Slope	Effective length of slope	Vertical above slope	Flat above slope	Route around side(s)

Position on diagram **			A	B	C	D		E	F	
FP denotes fish pass and EP Eel pass but measurements are for main structure only										
LILBURN										
V: Lilburn gravel trap (FP)	1.0		3.5	0.4	-	5.1%	2.0	-	-	No
W: Lilburn ford	0.5	to 0.7	-	0.5 <> 0.7 (c)	7.7	-	-	-	-	Yes
X: Lilburn bridge apron (FP)	0.4	to 0.9	-	0.20 to 0.5	-	-	-	-	10.0	No

(a) From substrate below structure to substrate above. At some structures, scour immediately downstream has created extra depth

(b) The structure is ruined but there is a natural steep slope

(c) Not absolutely vertical, there is a small ledge about 5cms from the top

Impassable at all water heights

impassable at low flows

Table 3.4 Assessment of Pool & Traverse type fish passes for lamprey passage (Photographs in Appendix C)

Pool & Traverse type STRUCTURE	Total height (a)	Height into entrance	Height of traverses (c)	No. of traverses	Overall slope	Effective Length	Status at visit
R TILL							
FP1: Twizel fish pass	1.5	0.31	0.2	2	na	na	Open
R. BREAMISH							
FP2: Powburn bridge fish pass	2.7	0	0.5	9*	12.0%	22.5	Open
WOOLER W.							

FP4: Wooler cauld fish pass	2	0.57	0.5	4	20.8%	9.6	Open
LILBURN							
FP7: Lilburn gravel trap fish pass	0.55	0	c. 0.3	3	9.3%	6.0	Open

Impassable at all water heights

Impassable at low flows

* with notches

(a)This is different from the total structure height when the fish pass exit is cut down in to it, but the same where this is not so

(b) Drop from bottom of pass to substrate below, but the downstream water level can be above the bottom of the pass

(c) Drop from top of one traverse to top of the next or, if notched, from base of notch in one traverse to base of notch in the next. Water levels will make actual jumps less than this

Table 3.5 Assessment of Larinier type fish passes for lamprey passage (Photographs in Appendix C)

Larinier type / Eel pass

STRUCTURE	Total height [a]	Vertical at entrance [b]	Slope	Effective length	Status at visit
R. BREAMISH					
FP3: Powburn bridge eel pass	2.7	0.0	14.3%	19	Blocked
WOOLER W.					
FP5: Wooler bridge fish pass	1.5	0.2	17.7%	7.0	Open
FP6: Haughhead ford fish pass [c]	2.4	0.0[d]	15.6%	5.84 [e]	Open
LILBURN					

FP8: Lilburn bridge fish pass	0.3	0	4.6%	6.7	Blocked

Impassable at all water heights	Impassable at low flows	Marginal

(a) Different from the total structure height when the fish pass exit is cut down in to it, but the same where this is not so

(b) Drop from bottom of pass to substrate below, but the downstream water level can be above the bottom of the pass

(c) The channel across the ford is 9.6m long of which the Larinier forms the first 5.8m from the downstream end. There is a 0.3 high steep incline in the channel 1.4m from the upstream end of the Larinier. The Slope and the Effective Length given are for the whole channel not just for the length of it that is formed by the Larinier

(d) There is a 0.4m drop from the base of the entrance of the Larinier pass to the substrate below, but the metal pilings downstream keep the water level up to the base so there can be no actual vertical drop, even in low flows (see Plate C13 in Appendix C)

(e) The Larinier is only part of a 9.6m long channel

REDD SURVEY

3.9 The sites visited to check for signs of spawning activity are mapped in Figure 3.3 and listed in Tables 3.6 and 3.7, with the dates on which they were checked. No certain signs of spawning activity of either River or Sea Lamprey were, however, seen.

3.10 However, a recollection of a pair of Lamprey "*about a yard long"* spawning at Tindal House on the lower Till about 15 years ago was collected from Brian Thompson, the boatman of Ford and Etal Estates. This is the first possible evidence of Sea Lamprey using the Till that has been found so far.

Figure 3.3 Locations of redd survey sites

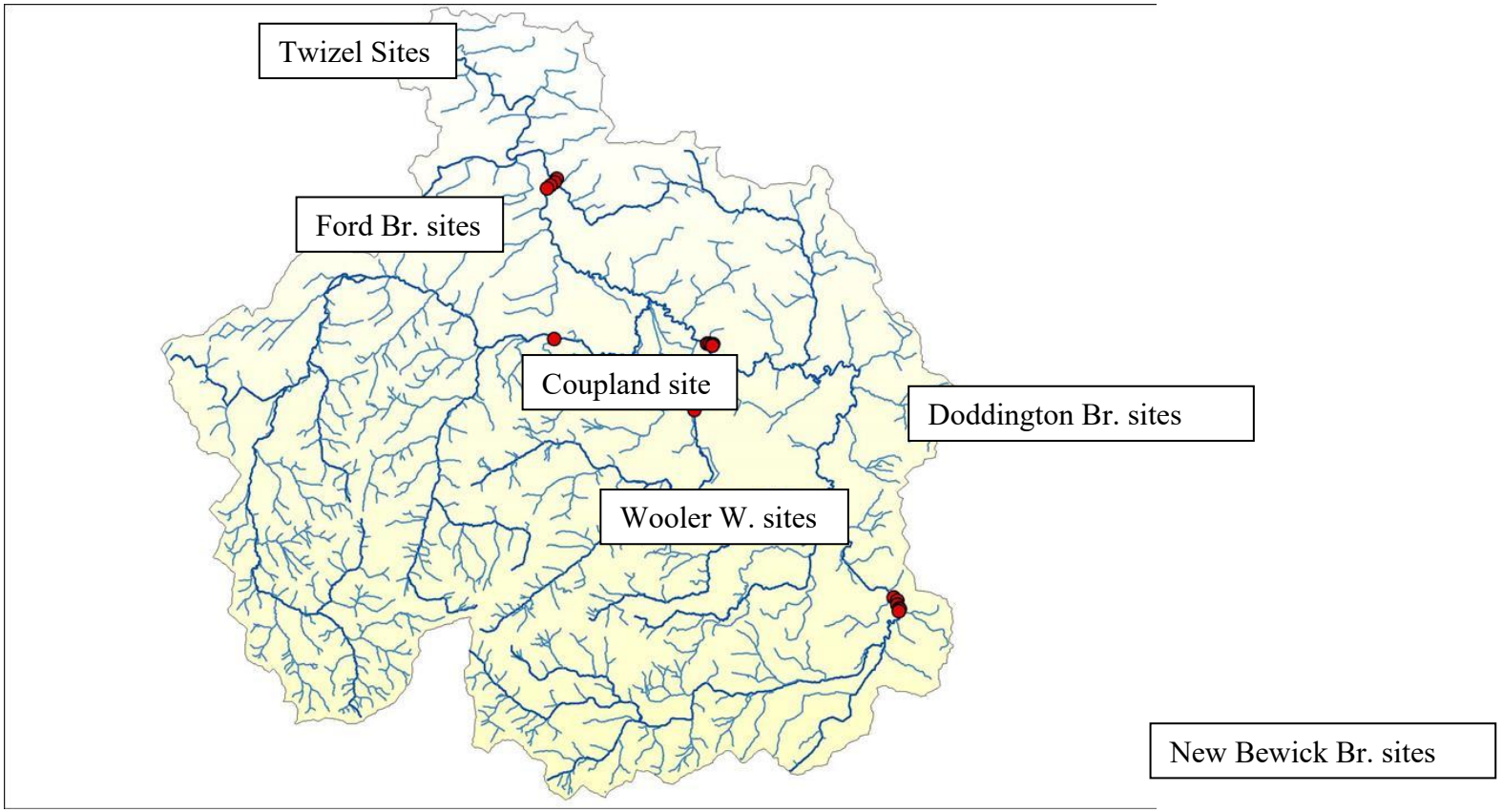


Table 3.6 Redd survey results from sites on the R. Till (Photographs in Appendix D)

Habitat Section	Ref. No.	Survey Site GPS	Date & Observations											
			APRIL			MAY		JUNE			JULY			
			23	27	30	12	19	3	9	17	30	10	15	
R. TILL														
T01	R01[a]	NT 87014 43012	0		0	X	0	X	0	0	0		0	
	R02	NT 88177 43209	0		0	0	0		0	0	0	0 [b]	0	
Till mouth to Twizel cauld	R03	NT 88302 43303	0		0	0	0		0	0	0		0	
	R04a	NT 88613 43018	0		0	0	0		0	0	0		0	
	R04b	NT 88654 42844	0		0	0	0		0	0	0		0	
	R04c	NT 88686 42700	0		0	0	0		0	0	0		0	
T02	R01	NT 93847 37485		A	0		0	X	0			X		
	R02	NT 93751 37337		0	0		0		0					
Upstream of Ford Bridge	R03	NT 93582 37212		A	0		0		0					
	R04	NT 93433 37086		0	0		0		0					
T04	R01	NU 00013 30720	0		0		0	X	0			X		
	R02	NU 00115 30726	0		0		0		0					
Upstream of Doddington Br	R03	NU 00207 30707	0		0		0		0					
	R04	NU 00246 30703	0		0		0		0					
	R05	NU 00221 30643	0		0		0		0					
T07	R01	NU 07658 20338	0		0		B		0					
	R02	NU 07802 20210	0		0		0		0					
Upstream of New Bewick (Harehope) Br	R03	NU 07804 20039	0		0		0		0					
	R04	NU 07882 19907	0		0		0		0					
	R05	NU 07901 19867			0		0		0					
	R06	NU 07870 19774	0		0		0		0					

A = possible redd
B = possible Brook Lamprey redd
X = water too coloured for visibility
(a) = Site in the Tweed, just off the mouth of the Till
(b) = possible to check when water too coloured elsewhere because shallow and close to bank

Table 3.7 Redd survey results from sites on the Wooler, Glen and Tweed (Photographs in Appendix D)

Habitat Section	Ref. No.	Survey Site GPS	Date & Observations										
			APRIL			MAY		JUNE			JULY		
			23	27	30	12	19	3	9	17	30	10	15
WOOLER WATER													
		NT 99427											
W01	R01	28546	0		0		0		0				
		NT 99519											
	R02	28615	0		0		0		0				
Below fish		NT 99564											
passes	R03	28698	B		0		0		0				
		NT 99838											
	R04	29581	0		0		0		0				
Between fish passes	R05		0		0		0	X	0			0	
R. GLEN													
G01 Coupland Ford	R01		0					0				0	

TWEED							
Downstream of Coldstream Br.	0		0	0	0	0	0
Upstream of Mertoun Br.	0		0			X	

A = possible

redd
B = possible Brook Lamprey redd
X = water too coloured for visibility
(a) = possible to check when water too coloured elsewhere because shallow and close to bank